

here: <https://dgit.in/MathMPy>. It is possible to import more than a single module in the same import statement. In such cases you will see that the module names will be separated by commas -

- `import math, random`

`import` statements can be positioned just about anywhere in a program, but it is considered good style to place it directly at

the very beginning of the program. In cases where only certain objects of a module are needed, you can go ahead and import only those specific objects -

- `from math import sin, pi`

Other objects such as `cos` will no longer be available after this import. But after this point, you can access `sin` and `pi` directly, without having to prefix them with “math”.

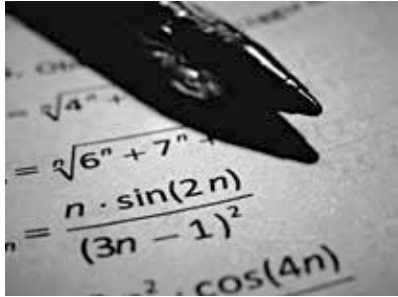
Instead of importing only certain objects from a module, you can import everything in the namespace of the importing module with an asterisk after the import statement as shown here: <https://dgit.in/ImpPy>.

It is not usually recommended to use the asterisk notation except for while working in the interactive Python shell. One of the simple reasons for it is that the origin of a name could, in some cases, be obscure, because you cannot see which module it may have been imported from.

People usually use the asterisk notation because of the simple fact that it is convenient. Using the asterisk means avoiding a lot of tedious typing. Another simple method that can be utilized to shrink the typing effort is by renaming a namespace. A fabulous and often used example for this is the `numpy` module. It is extremely rare to find an example in any Python tutorial where you see this module being imported with the following statement -

- `import numpy`. It is almost as if it is an unwritten constitutional law to import it with the following statement: `import numpy as np`.

From this point on, you can prefix all of the objects of `numpy` with just “`np`” as opposed to “`numpy`”, as shown here: <https://dgit.in/RenamePy>.



Understanding modules is simplified by using the example of a math module

Designing Modules

So now you know what modules are, but how does one actually create modules? A module in Python is a file containing definitions and statements (as